COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9/17/26	**DATE SUBMITTED:**	9/19/26
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ACTIVITY NO. 8.1 **Install, Configure, and Manage Enterprise Availability Monitoring via Ansible**

1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

System Setup

VM	OS
Control Node	**Ubuntu**
Managed Node 1	**Ubuntu**
Managed Node 2	**Ubuntu**
Managed Node 3	**CentOS**

Network Config

VM	IP
Control Node	**192.168.56.107**
Managed Node 1	**192.162.56.109**
Managed Node 2	**192.168.56.108**
Managed Node 3	**192.162.56.110**

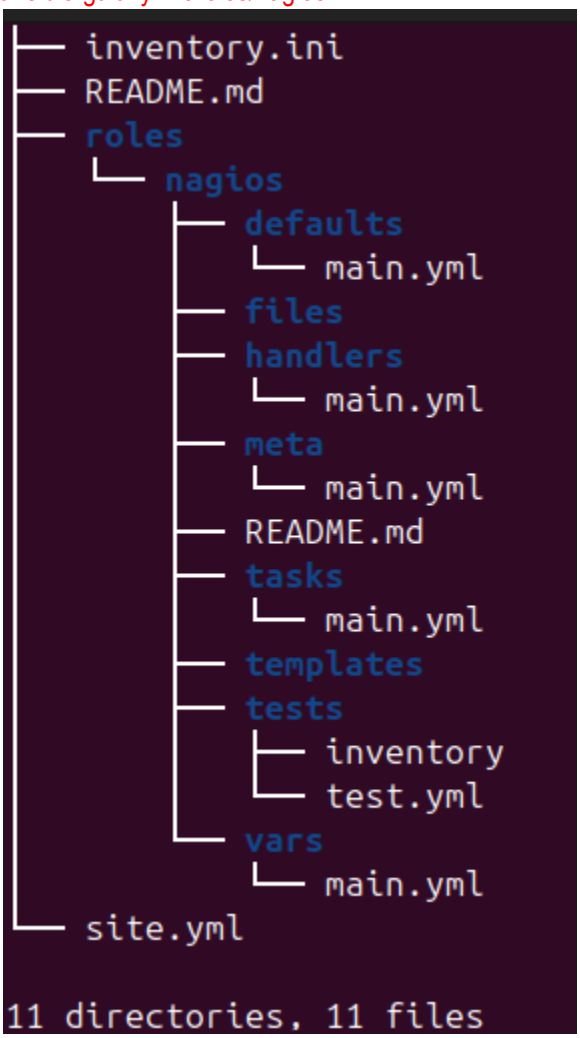
Pre-requisites

Control Node

```
sudo apt update
sudo apt install ansible -y
```

Create and separate directory for the activity
mkdir ~/HOA8
cd ~/HOA8

Application of creating roles
ansible-galaxy init roles/nagios



Config inventory.ini

```
  GNU nano 7.2                                inventory.ini
[ubuntu]
ubuntu1 ansible_host=192.168.56.109
ubuntu2 ansible_host=192.168.56.108

[centos]
centos1 ansible_host=192.168.56.110

[all:vars]
ansible_user=villareal
```

Create main playbook (site.yml)

nano site.yml

```
  GNU nano 7.2                                 site.yml
---
- name: Install Nagios on managed nodes
  hosts: all
  become: true

  roles:
    - nagios
```

Configuring nagios role for each managed nodes

nano roles/nagios/tasks/main.yml

GNU nano 7.2 roles/nagios/tasks/main.yml *

```
- name: Update ubuntu repository
  ansible.builtin.apt:
    update_cache: yes
  when: ansible_os_family == "Debian"

- name: Install ubuntu dependecies
  ansible.builtin.apt:
    name:
      - apache2
      - apache2-utils
      - php
      - libapache2-mod-php
      - php-gd
      - build-essential
      - autoconf
      - gcc
      - make
      - wget
      - unzip
      - libgd-dev
      - libssl-dev
    state: present
  when: ansible_os_family == "Debian"
```

```
  GNU nano 7.2                    roles/nagios/tasks/main.yml *
- name: install DNF plugins
  ansible.builtin.dnf:
    name: dnf-plugins-core
    state: present
  when: ansible_os_family == "RedHat"

- name: enable appstream repository
  ansible.builtin.command:
    cmd: dnf config-manager --set-enabled appstream
  when: ansible_os_family == "RedHat"
  changed_when: false
  failed_when: false

- name: Enable CRB repository
  ansible.builtin.command:
    cmd: dnf config-manager --set-enabled crb
  when:
    - ansible_os_family == "RedHat"
    - ansible_distribution_major_version | int >= 9
  changed_when: false
  failed_when: false

- name: Update centos repository
  ansible.builtin.dnf:
    update_cache: yes
  when: ansible_os_family == "RedHat"
```

```
  GNU nano 7.2                    roles/nagios/tasks/main.yml *
- name: Install CentOS dependencies
  ansible.builtin.dnf:
    name:
      - httpd
      - httpd-tools
      - php
      - gcc
      - glibc
      - glibc-common
      - perl
      - gd
      - gd-devel
      - autoconf
      - make
      - wget
      - unzip
      - openssl-devel
    state: present
    update_cache: yes
  when: ansible_os_family == "RedHat"

- name: Create nagios group
  ansible.builtin.group:
    name: nagios
    state: present
```

GNU nano 7.2 roles/nagios/tasks/main.yml *

```
- name: Create nagios user
  ansible.builtin.user:
    name: nagios
    group: nagios
    state: present

- name: Create nagcmd group
  ansible.builtin.group:
    name: nagcmd
    state: present

- name: Add nagios user to nagcmd
  ansible.builtin.user:
    name: nagcmd
    state: present

- name: Add apache user to nagcmd in ubuntu
  ansible.builtin.user:
    name: www-data
    groups: nagcmd
    append: yes
  when: ansible_os_family == "Debian"

- name: Add apache user to nagcmd on centos
  ansible.builtin.user:
    name: apache
    groups: nagcmd
```

```
    append: yes
  when: ansible_os_family == "RedHat"

- name: Download nagios core
  ansible.builtin.get_url:
    url: https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.14
    dest: /tmp/nagios-4.5.14.tar.gz
    mode: '0644'

- name: Extract Nagios core
  ansible.builtin.unarchive:
    src: /tmp/nagios-4.5.14.tar.gz
    dest: /tmp
    remote_src: yes

- name: configure nagios core
  ansible.builtin.command:
    cmd: ./configure --with-command-group=nagcmd
    chdir: /tmp/nagios-4.5.14
  args:
    creates: /tmp/nagios-4.5.14/Makefile

- name: Compile nagios core
  ansible.builtin.command:
    cmd: make all
    chdir: /tmp/nagios-4.5.14
```

```
  GNU nano 7.2                  roles/nagios/tasks/main.yml *
- name: Install nagios core
  ansible.builtin.command:
    cmd: make install
    chdir: /tmp/nagios-4.5.14

- name: Install nagios configuration
  ansible.builtin.command:
    cmd: make install-config
    chdir: /tmp/nagios-4.5.14

- name: install nagios command mode
  ansible.builtin.command:
    cmd: make install-commandmode
    chdir: /tmp/nagios-4.5.14

- name: install nagios service
  ansible.builtin.command:
    cmd: make install-daemoninit
    chdir: /tmp/nagios-4.5.14

- name: remove existing nagios web config
  ansible.builtin.file:
    path: /etc/apache2/sites-enabled/nagios.conf
    state: absent
  when: ansible_os_family == "Debian"

- name: install nagios webconfiguration
  ansible.builtin.command:
```

```
    cmd: make install-webconf
    chdir: /tmp/nagios-4.5.14
  when: ansible_os_family == "Debian"

- name: restart apache on ubuntu
  ansible.builtin.service:
    name: apache2
    state: restarted
    enabled: yes
  when: ansible_os_family == "Debian"

- name: restart apache on centos
  ansible.builtin.service:
    name: httpd
    state: restarted
    enabled: yes
  when: ansible_os_family == "RedHat"

- name: create nagios web authetication on ubuntu
  ansible.builtin.command:
    cmd: htpasswd -b -c /usr/local/nagios/etc/htpasswd.users nagiosadmin nagio>
    creates: /usr/local/nagios/etc/htpasswd.users
  when: ansible_os_family == "Debian"
```

```
- name: create nagios web authentication on centos
  ansible.builtin.command:
    cmd: htpasswd -b -c /usr/local/nagios/etc/htpasswd.users nagiosadmin nagio>
    creates: /usr/local/nagios/etc/htpasswd.users
  when: ansible_os_family == "RedHat"

- name: Download nagios plugins
  ansible.builtin.get_url:
    url: https://github.com/nagios-plugins/nagios-plugins/releases/download/re>
    dest: /tmp/nagios-plugins-2.5.tar.gz
    mode: '0644'

- name: Extract nagios plugins
  ansible.builtin.unarchive:
    src: /tmp/nagios-plugins-2.5.tar.gz
    dest: /tmp
    remote_src: yes

- name: Configure nagios plugins
  ansible.builtin.command:
    cmd: ./configure --with-nagios-user=nagios --with-nagios-group=nagios
    chdir: /tmp/nagios-plugins-2.5
  args:
    creates: /tmp/nagios-plugins-2.5/Makefile

- name: Compile nagios plugins
  ansible.builtin.command:
    cmd: make
```

```
    chdir: /tmp/nagios-plugins-2.5

- name: Install nagios plugins
  ansible.builtin.command:
    cmd: make install
    chdir: /tmp/nagios-plugins-2.5

- name: Validate nagios configuration
  ansible.builtin.command:
    cmd: /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
  register: nagios_config_check
  changed_when: false

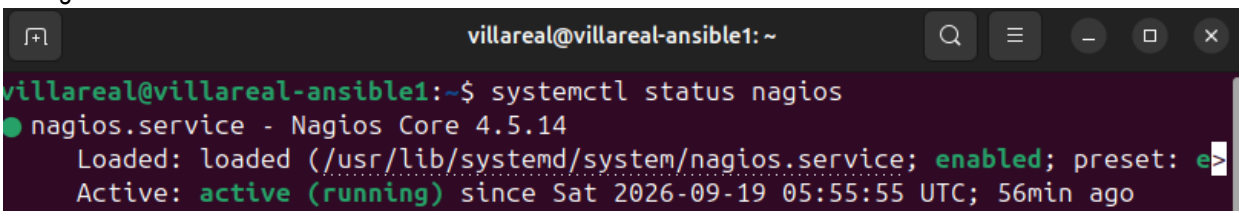
- name: Start and enable nagios
  ansible.builtin.service:
    name: nagios
    state: started
    enabled: yes
```

Run the playbook
ansible-playbook -i inventory.ini site.yml –ask-become-pass

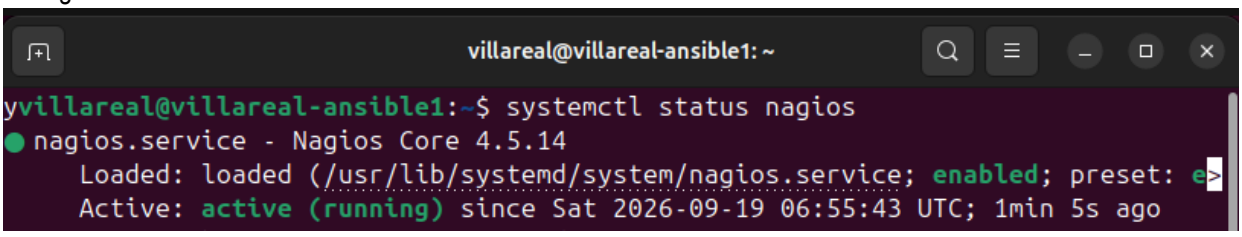
```
PLAY RECAP *********************************************************************
centos1                    : ok=28   changed=22   unreachable=0    failed=0    s
kipped=9    rescued=0    ignored=0
ubuntu1                    : ok=29   changed=13   unreachable=0    failed=0    s
kipped=8    rescued=0    ignored=0
ubuntu2                    : ok=29   changed=13   unreachable=0    failed=0    s
kipped=8    rescued=0    ignored=0
```

Verify that nagios is installed on the managed nodes
systemctl status nagios

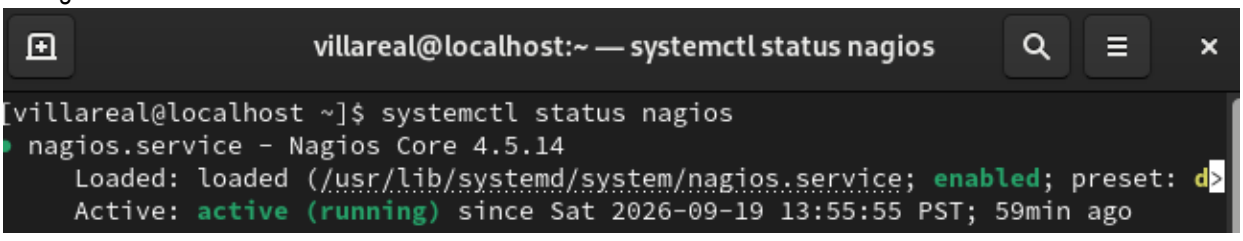
Managed node 1



Managed node 2

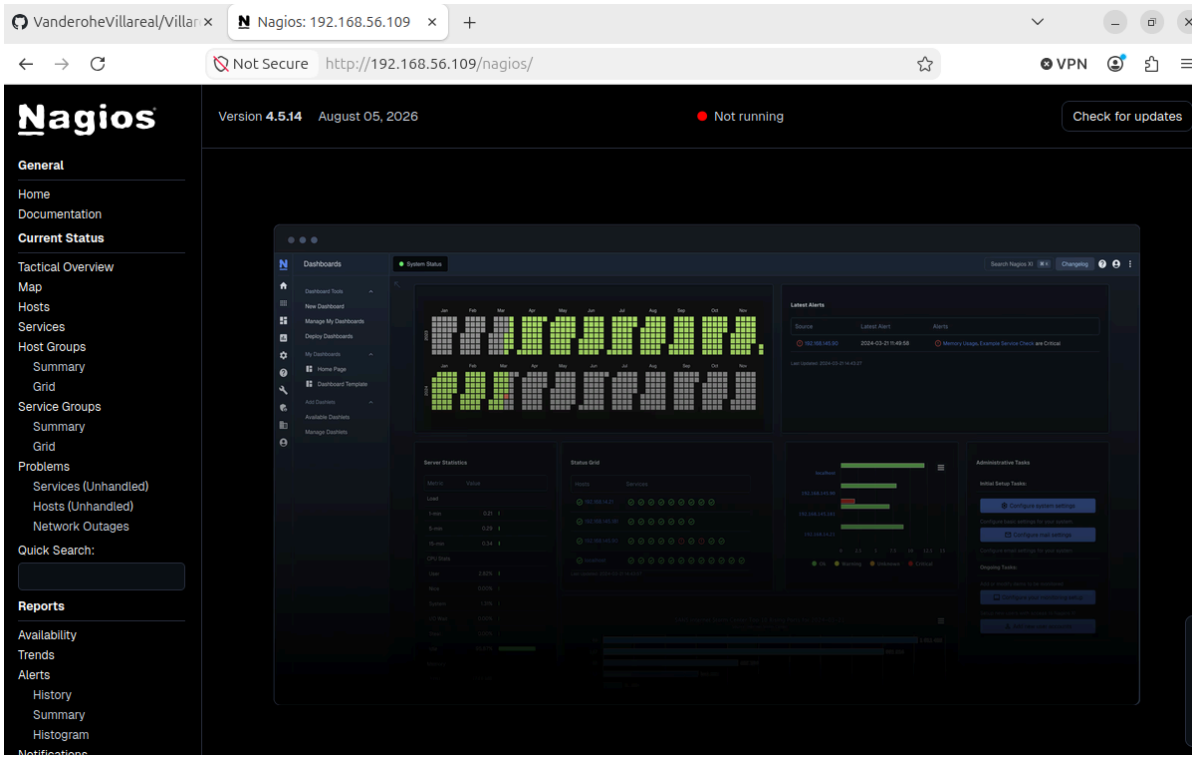


Managed Node 3

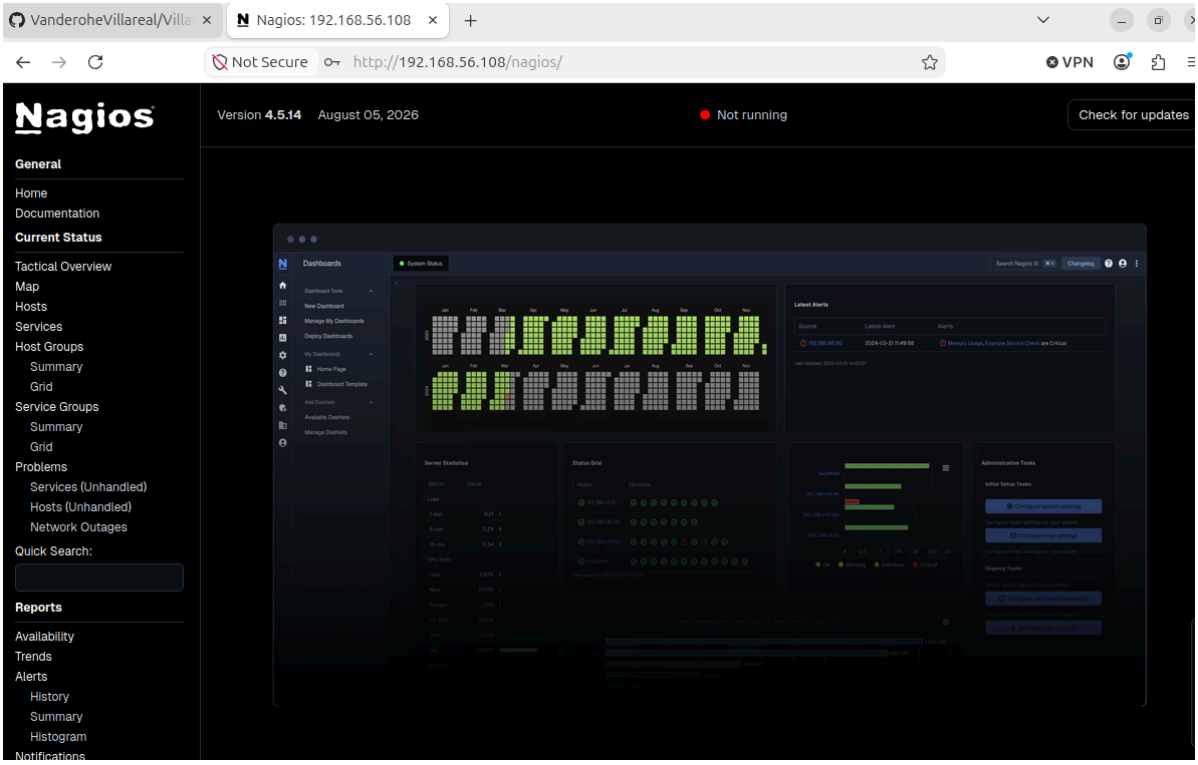


Verification on the web interface
Search http://<IP-ADD>/nagios

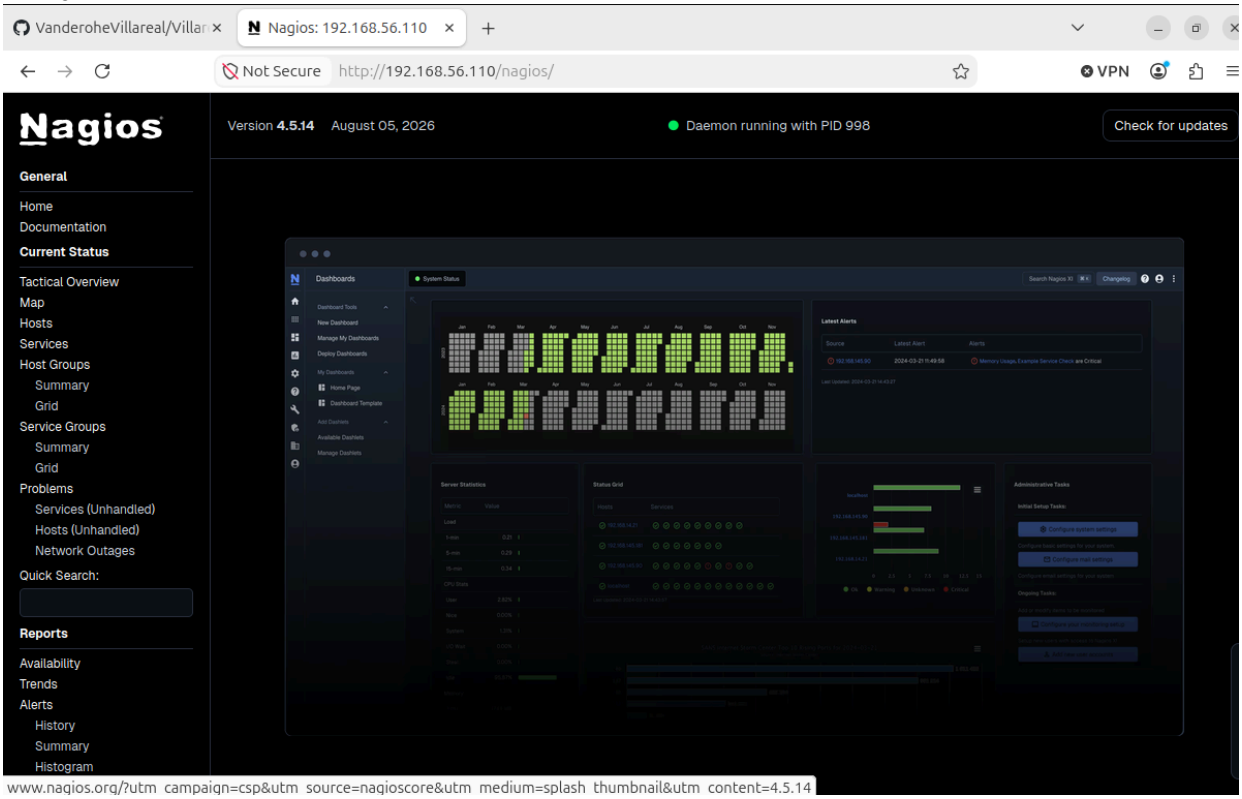
Managed node 1



Managed node 2



Managed Node 3



Troubleshooting:

Problem 1:
gd-devel unavailable, on this problem I encountered errors that keep happening when running the playbook.

```
TASK [nagios : Install CentOS dependencies] ************************************
*
skipping: [ubuntu1]
skipping: [ubuntu2]
fatal: [centos1]: FAILED! => {"changed": false, "failures": ["No package gd-lev
el available."], "msg": "Failed to install some of the specified packages", "rc
": 1, "results": []}
```

Problem was in the main.yml file where I mistyped gd-devel as gd-level

Problem 2:
Update_cache unsupported, this was also a problem I encountered

```
TASK [nagios : Update centos repository] ***********************************
*
skipping: [ubuntu1]
skipping: [ubuntu2]
fatal: [centos1]: FAILED! => {"changed": false, "msg": "Unsupported parameters
for (ansible.legacy.command) module: update_cache. Supported parameters include
: _raw_params, _uses_shell, argv, chdir, creates, executable, expand_argument_v
ars, removes, stdin, stdin_add_newline, strip_empty_ends."}
```

Same mistake as the first problem mistyped update_cache as updated_cache

Problem 3:
Nagios Plugins 404, error shows HTTP Error 404: Not Found

```
TASK [nagios : Download nagios plugins] ************************************
*
fatal: [ubuntu1]: FAILED! => {"changed": false, "dest": "/tmp/nagios-plugins-2.
5.tar.gz", "elapsed": 0, "msg": "Request failed", "response": "HTTP Error 404:
Not Found", "status_code": 404, "url": "https://github.com/nagios-plugins/nagio
s-plugins/releases/download/releases/download/release-2.5/nagios-plugins-2.5.ta
r.gz"}
fatal: [ubuntu2]: FAILED! => {"changed": false, "dest": "/tmp/nagios-plugins-2.
5.tar.gz", "elapsed": 0, "msg": "Request failed", "response": "HTTP Error 404:
Not Found", "status_code": 404, "url": "https://github.com/nagios-plugins/nagio
s-plugins/releases/download/releases/download/release-2.5/nagios-plugins-2.5.ta
r.gz"}
```

Same mistake, mistyped the github link inside the main.yml file therefore caused an error which led to the failed task.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?

Availability monitoring tools help detect downtime or service problems quickly, allowing administrators to respond immediately, reduce downtime, and keep systems reliable and accessible.

Conclusions/Learnings:

The Nagios Core installation was successfully automated using Ansible Roles on both Ubuntu and CentOS Stream 9 managed nodes. The role handled the installation of dependencies, Nagios Core, Nagios Plugins, Apache configuration, authentication, and service management. The successful playbook execution and service validation confirmed that the automation worked correctly across both operating systems.

GitHub Link:

https://github.com/VanderoheVillareal/Villareal_NAGIOS

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error encountered**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI is used to help me in resolving and identifying errors during the entire session of this activity.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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